



Brief paper

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ABSTRACT

Traditional Kalman filter (KF) is derived under the well-known minimum mean square error (MMSE) criterion, which is optimal under Gaussian assumption. However, when the signals are non-Gaussian, especially when the system is disturbed by some heavy-tailed impulsive noises, the performance of KF will deteriorate seriously. To improve the robustness of KF against impulsive noises, we propose in this work a new Kalman filter, called the maximum correntropy Kalman filter (MCKF), which adopts the robust maximum correntropy criterion (MCC) as the optimality criterion, instead of using the MMSE. Similar to the traditional KF, the state mean vector and covariance matrix propagation equations are used to give prior estimations of the state and covariance matrix in MCKF. A novel fixed-point algorithm is then used to update the posterior estimations. A sufficient condition that guarantees the convergence of the fixed-point algorithm is also given. Illustration examples are presented to demonstrate the effectiveness and robustness of the new algorithm.

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1. Introduction

Estimation problem has been one of the most important issues from industrial appliances to research areas including signal processing, optimal control, navigation and so on. The actual applications include parameter estimate (Tambini, Montanari, & Cacciari, 1992), system identification (Unsal & Dogan, 2014), target tracking (Yadav, Naik, Ananthasayanam, Gaur, & Singh, 2012), simultaneous localization (Chou, Traonmilin, Ollivier, & Parent, 2004) and many others. For linear dynamic systems, the estimation problem is usually solved by Kalman filter (KF), which is, in essence, a least squares linear filter that provides an optimal recursive solution (Bryson & Ho, 1975; Kalman, 1960; Nahi, 1969). In order to solve the nonlinear filtering problems, researchers have made many additions to the Kalman family filters, including

extended Kalman filter (EKF), Second-order extended Kalman filter (SEKF), unscented Kalman filter (UKF), cubature Kalman filter (CKF) and many others. EKF (Anderson & Moore, 1979) is a basic extension to the KF, which approximates the nonlinear system by its first order linearization. SEKF (Roth & Gustafsson, 2011) maintains the second-order terms of the Taylor expansions of the state and measurement equations. UKF (Julier, Uhlmann, & Durrant-Whyte, 2000) was derived based on the unscented transform for approximation, and CKF (Arasaratnam & Haykin, 2009) was developed by using a spherical-radial cubature rule for approximating the Gaussian filter. At the same time, robust Kalman filters were also studied. Examples include the fixed-lag doubly robust smoothing (DRS) (Farahmand, Giannakis, & Angelosante, 2011), sum-of-norms regularization (Ohlsson, Gustafsson, Ljung, & Boyd, 2012), non-smooth quadratic support (QS) penalties (Aravkin, Burke, & Pillonetto, 2013) and so on. Moreover, H_∞ filters (Gao & Li, 2011; Li & Gao, 2012; Li, Lam, Gao, & Xiong, 2016; Xie, Souza, & Fu, 1991) are robust filters that can solve efficiently the model uncertainty problems. The particle filter (PF) (Arulampalam, Maskell, Gordon, & Clapp, 2002) is also a robust filter, which is a sequential Monte Carlo method based on the particle representations of probability distributions.

The KF and its extensions in general perform very well in Gaussian noises (Morris, 1976). Nevertheless, their performances are likely to get worse when applied to non-Gaussian situations, especially when the systems are disturbed by impulsive noises.

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The impulsive noises are of heavy-tailed distributions (such as some mixed-Gaussian distributions), which are common in many real scenarios of automatic control and target tracking. The main reason for this problem is that the KF is based on the well-known minimum mean square error (MMSE) criterion, which is sensitive to large outliers and results in deterioration of the robustness of the KF in non-Gaussian noise environments (Wu, Shi, Zhang, Ma, & Chen, 2015).

The optimization criteria in information theoretic learning (ITL) (Chen, Zhu, Hu, & Principe, 2013; Principe, 2010) have gained increasing attention over the past few years, which use the information theoretic quantities (e.g. entropy) estimated directly from the data instead of the usual second order statistical measures, such as variance and covariance, as the optimization costs. Information theoretic quantities can capture higher-order statistics and offer potentially significant performance improvement in machine learning and signal processing applications. The ITL links information theory, nonparametric estimators, and reproducing kernel Hilbert spaces (RKHS) in a simple and unconventional way. In particular, the correntropy as a nonlinear similarity measure in kernel space has its root in Renyi's entropy (Chen & Principe, 2012; Chen, Xing, Liang, Zheng, & Principe, 2014; Liu, Pokharel, & Principe, 2007). Since correntropy is also a local similarity measure (hence insensitive to outliers), it is naturally a robust cost for machine learning and signal processing (He, Hu, Zheng, & Kong, 2011; He, Zheng, & Hu, 2011). In supervised learning, such as regression, the problem can be formulated as that of maximizing the correntropy between model output and desired response. This optimization criterion is called in ITL the maximum correntropy criterion (MCC) (Chen et al., 2013; Principe, 2010). Recently, the MCC has been successfully used in robust adaptive filtering in impulsive (heavy-tailed) noise environments (Chen et al., 2014; Chen, Xing, Zhao, Zheng, & Principe, 2016; Principe, 2010).

The MCC solution cannot be obtained in closed form even for a simple linear regression problem, so one has to solve it using an iterative update algorithm such as the gradient based methods (Chen et al., 2014). The gradient based methods are simple and widely used. But they depend on a free parameter step-size and usually converge to an optimal solution slowly. The fixed-point iterative algorithm is an alternative efficient way to solve the MCC solution, which involves no step-size and may converge to the solution very fast (Agarwal, Meehan, & Regan, 2001; Principe, 2010; Singh & Principe, 2010). A sufficient condition that guarantees the convergence of the fixed-point MCC algorithm was given in Chen, Wang, Zhao, Zheng, and Principe (2015).

In the present paper, we develop a new Kalman filter, compactly called the maximum correntropy Kalman filter (MCKF), based on the MCC and a fixed-point iterative algorithm. Similar to the traditional KF, the MCKF not only retains the state mean propagation process, but also preserves the covariance matrix propagation process. Thus the new filter also has a recursive structure and is suitable for online implementation. It is worth noting that in Cinar and Principe (2012), the MCC has been used in hidden state estimation, but it involves no covariance propagation process and is in form not a Kalman filter. In this work, for simplicity our focus is mainly on the linear model. The nonlinear extensions of MCKF are left for future study.

The rest of the paper is organized as follows. In Section 2, we briefly introduce the correntropy and Kalman filter. In Section 3, we derive the MCKF algorithm and present the computational complexity and convergence analysis results. Illustrative examples are then provided in Section 4 to show the excellent performance of the MCKF. Finally, conclusion is given in Section 5.

2. Preliminaries

2.1. Correntropy

Correntropy is a generalized similarity measure between two random variables. Given two random variables $X, Y \in \mathbb{R}$ with joint distribution function $F_{XY}(x, y)$, correntropy is defined by

$$V(X, Y) = E[\kappa(X, Y)] = \int \kappa(x, y) dF_{XY}(x, y) \quad (1)$$

where E denotes the expectation operator, and $\kappa(\cdot, \cdot)$ is a shift-invariant Mercer Kernel. In this paper, without mentioned otherwise the kernel function is the Gaussian Kernel given by

$$\kappa(x, y) = G_\sigma(e) = \exp\left(-\frac{e^2}{2\sigma^2}\right) \quad (2)$$

where $e = x - y$, and $\sigma > 0$ stands for the kernel bandwidth.

In most practical situations, however, only limited numbers of data are available and the joint distribution F_{XY} is usually unknown. In these cases, one can estimate the correntropy using a sample mean estimator:

$$\hat{V}(X, Y) = \frac{1}{N} \sum_{i=1}^N G_\sigma(e(i)) \quad (3)$$

where $e(i) = x(i) - y(i)$, with $\{x(i), y(i)\}_{i=1}^N$ being N samples drawn from F_{XY} .

Taking Taylor series expansion of the Gaussian Kernel, we have

$$V(X, Y) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n \sigma^{2n} n!} E[(X - Y)^{2n}]. \quad (4)$$

As one can see, the correntropy is a weighted sum of all even order moments of the random variable $X - Y$. The kernel bandwidth appears as a parameter weighting the second order and higher order moments. With a very large σ (compared to the dynamic range of the data), the correntropy will be dominated by the second order moment.

2.2. Kalman filter

Kalman filter provides a powerful tool to deal with state estimation of linear systems, which is an optimal estimator under linear and Gaussian assumptions.

Consider a linear system described by the following state and measurement equations:

$$\mathbf{x}(k) = \mathbf{F}(k-1)\mathbf{x}(k-1) + \mathbf{q}(k-1), \quad (5)$$

$$\mathbf{y}(k) = \mathbf{H}(k)\mathbf{x}(k) + \mathbf{r}(k) \quad (6)$$

where $\mathbf{x}(k) \in \mathbb{R}^n$ denotes the n -dimensional state vector, $\mathbf{y}(k) \in \mathbb{R}^m$ represents the m -dimensional measurement vector at instant k . $\mathbf{F}$ and $\mathbf{H}$ stand for, respectively, the system matrix (or state transition matrix) and observation matrix. $\mathbf{q}(k-1)$ and $\mathbf{r}(k)$ are mutually uncorrelated process noise and measurement noise, respectively, with zero mean and covariance matrices

$$E[\mathbf{q}(k-1)\mathbf{q}^T(k-1)] = \mathbf{Q}(k-1), \quad E[\mathbf{r}(k)\mathbf{r}^T(k)] = \mathbf{R}(k). \quad (7)$$

In general, Kalman filter includes the following two steps:

(1) Predict: The prior mean and covariance matrix are given by

$$\hat{\mathbf{x}}(k|k-1) = \mathbf{F}(k-1)\hat{\mathbf{x}}(k-1|k-1), \quad (8)$$

$$\mathbf{P}(k|k-1) = \mathbf{F}(k-1)\mathbf{P}(k-1|k-1)\mathbf{F}^T(k-1) + \mathbf{Q}(k-1). \quad (9)$$

(2) Update: The Kalman filter gain is computed as

$$\mathbf{K}(k) = \mathbf{P}(k|k-1)\mathbf{H}^T(k) (\mathbf{H}(k)\mathbf{P}(k|k-1)\mathbf{H}^T(k) + \mathbf{R}(k))^{-1}. \quad (10)$$

The posterior state is equal to the prior state plus the innovation weighted by the KF gain,

$$\hat{\mathbf{x}}(k|k) = \hat{\mathbf{x}}(k|k-1) + \mathbf{K}(k) (\mathbf{y}(k) - \mathbf{H}(k)\hat{\mathbf{x}}(k|k-1)). \quad (11)$$

Additionally, the posterior covariance is recursively updated as follows:

$$\mathbf{P}(k|k) = (\mathbf{I} - \mathbf{K}(k)\mathbf{H}(k)) \mathbf{P}(k|k-1) (\mathbf{I} - \mathbf{K}(k)\mathbf{H}(k))^T + \mathbf{K}(k)\mathbf{R}(k)\mathbf{K}^T(k). \quad (12)$$

3. Kalman filter under MCC

3.1. Derivation of the algorithm

Traditional Kalman filter works well under Gaussian noises, but its performance may deteriorate significantly under non-Gaussian noises, especially when the underlying system is disturbed by impulsive noises. The main reason for this is that the KF is developed based on the MMSE criterion, which captures only the second order statistics of the error signal and is sensitive to large outliers. To address this problem, in this work we use the maximum correntropy criterion (MCC) to derive a new Kalman filter, which may perform much better in non-Gaussian noise environments, as correntropy contains second and higher order moments of the error.

For the linear model described in the previous section, we have

$$\begin{bmatrix} \hat{\mathbf{x}}(k|k-1) \\ \mathbf{y}(k) \end{bmatrix} = \begin{bmatrix} \mathbf{I} \\ \mathbf{H}(k) \end{bmatrix} \mathbf{x}(k) + \nu(k) \quad (13)$$

where $\mathbf{I}$ is the $n \times n$ identity matrix, and $\nu(k)$ is

$$\nu(k) = \begin{bmatrix} -(\mathbf{x}(k) - \hat{\mathbf{x}}(k|k-1)) \\ \mathbf{r}(k) \end{bmatrix},$$

with

$$\begin{aligned} \mathbb{E}[\nu(k)\nu^T(k)] &= \begin{bmatrix} \mathbf{P}(k|k-1) & 0 \\ 0 & \mathbf{R}(k) \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{B}_p(k|k-1)\mathbf{B}_p^T(k|k-1) & 0 \\ 0 & \mathbf{B}_r(k)\mathbf{B}_r^T(k) \end{bmatrix} \\ &= \mathbf{B}(k)\mathbf{B}^T(k) \end{aligned} \quad (14)$$

where $\mathbf{B}(k)$ can be obtained by Cholesky decomposition of $\mathbb{E}[\nu(k)\nu^T(k)]$. Left multiplying both sides of (13) by $\mathbf{B}^{-1}(k)$, we get

$$\mathbf{D}(k) = \mathbf{W}(k)\mathbf{x}(k) + \mathbf{e}(k) \quad (15)$$

where $\mathbf{D}(k) = \mathbf{B}^{-1}(k) \begin{bmatrix} \hat{\mathbf{x}}(k|k-1) \\ \mathbf{y}(k) \end{bmatrix}$, $\mathbf{W}(k) = \mathbf{B}^{-1}(k) \begin{bmatrix} \mathbf{I} \\ \mathbf{H}(k) \end{bmatrix}$, $\mathbf{e}(k) = \mathbf{B}^{-1}(k)\nu(k)$. Since $\mathbb{E}[\mathbf{e}(k)\mathbf{e}^T(k)] = \mathbf{I}$, the residual error $\mathbf{e}(k)$ is white.

Now we propose the following MCC based cost function:

$$J_L(\mathbf{x}(k)) = \frac{1}{L} \sum_{i=1}^L G_\sigma(d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k)) \quad (16)$$

where $d_i(k)$ is the i th element of $\mathbf{D}(k)$, $\mathbf{w}_i(k)$ is the i th row of $\mathbf{W}(k)$, and $L = n + m$ is the dimension of $\mathbf{D}(k)$. Then, under MCC, the optimal estimate of $\mathbf{x}(k)$ is

$$\hat{\mathbf{x}}(k) = \arg \max_{\mathbf{x}(k)} J_L(\mathbf{x}(k)) = \arg \max_{\mathbf{x}(k)} \sum_{i=1}^L G_\sigma(e_i(k)) \quad (17)$$

where $e_i(k)$ is the i th element of $\mathbf{e}(k)$:

$$e_i(k) = d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k). \quad (18)$$

The optimal solution can thus be obtained by solving

$$\begin{aligned} \frac{\partial J_L(\mathbf{x}(k))}{\partial \mathbf{x}(k)} &= \sum_{i=1}^L [G_\sigma(e_i(k)) \mathbf{w}_i^T(k) (d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k))] = 0. \end{aligned} \quad (19)$$

It follows easily that

$$\begin{aligned} \mathbf{x}(k) &= \left(\sum_{i=1}^L [G_\sigma(e_i(k)) \mathbf{w}_i^T(k) \mathbf{w}_i(k)] \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^L [G_\sigma(e_i(k)) \mathbf{w}_i^T(k) d_i(k)] \right). \end{aligned} \quad (20)$$

Since $e_i(k) = d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k)$, the optimal solution (20) is actually a fixed-point equation of $\mathbf{x}(k)$ and can be rewritten as

$$\mathbf{x}(k) = \mathbf{f}(\mathbf{x}(k)) \quad (21)$$

with

$$\begin{aligned} \mathbf{f}(\mathbf{x}(k)) &= \left(\sum_{i=1}^L [G_\sigma(d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k)) \mathbf{w}_i^T(k) \mathbf{w}_i(k)] \right)^{-1} \\ &\quad \times \left(\sum_{i=1}^L [G_\sigma(d_i(k) - \mathbf{w}_i(k)\mathbf{x}(k)) \mathbf{w}_i^T(k) d_i(k)] \right). \end{aligned}$$

A fixed-point iterative algorithm can be readily obtained as

$$\hat{\mathbf{x}}(k)_{t+1} = \mathbf{f}(\hat{\mathbf{x}}(k)_t) \quad (22)$$

where $\hat{\mathbf{x}}(k)_t$ denotes the solution at the fixed-point iteration t .

The fixed-point equation (20) can also be expressed as

$$\mathbf{x}(k) = (\mathbf{W}^T(k)\mathbf{C}(k)\mathbf{W}(k))^{-1} \mathbf{W}^T(k)\mathbf{C}(k)\mathbf{D}(k) \quad (23)$$

where $\mathbf{C}(k) = \begin{bmatrix} \mathbf{C}_x(k) & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_y(k) \end{bmatrix}$, with

$$\mathbf{C}_x(k) = \text{diag}(G_\sigma(e_1(k)), \dots, G_\sigma(e_n(k))),$$

$$\mathbf{C}_y(k) = \text{diag}(G_\sigma(e_{n+1}(k)), \dots, G_\sigma(e_{n+m}(k))).$$

Eq. (23) can be further expressed as follows (see Appendix for a detailed derivation):

$$\mathbf{x}(k) = \hat{\mathbf{x}}(k|k-1) + \bar{\mathbf{K}}(k) (\mathbf{y}(k) - \mathbf{H}(k)\hat{\mathbf{x}}(k|k-1)) \quad (24)$$

where

$$\begin{cases} \bar{\mathbf{K}}(k) = \bar{\mathbf{P}}(k|k-1) \mathbf{H}^T(k) (\mathbf{H}(k) \bar{\mathbf{P}}(k|k-1) \mathbf{H}^T(k) + \bar{\mathbf{R}}(k))^{-1} \\ \bar{\mathbf{P}}(k|k-1) = \mathbf{B}_p(k|k-1) \mathbf{C}_x^{-1}(k) \mathbf{B}_p^T(k|k-1) \\ \bar{\mathbf{R}}(k) = \mathbf{B}_r(k) \mathbf{C}_y^{-1}(k) \mathbf{B}_r^T(k). \end{cases} \quad (25)$$

Remark 1. Of course, Eq. (24) is also a fixed-point equation of $\mathbf{x}(k)$ because $\bar{\mathbf{K}}(k)$ depends on $\bar{\mathbf{P}}(k|k-1)$ and $\bar{\mathbf{R}}(k)$, both related to $\mathbf{x}(k)$ via $\mathbf{C}_x(k)$ and $\mathbf{C}_y(k)$, respectively. The optimal solution of Eq. (24) depends also on the prior estimate $\hat{\mathbf{x}}(k|k-1)$, which can be calculated by (8) using the latest estimate $\hat{\mathbf{x}}(k-1|k-1)$.

With the above derivations, we summarize the proposed MCKF algorithm as follows:

- (1) Choose a proper kernel bandwidth σ and a small positive number ε ; Set an initial estimate $\hat{\mathbf{x}}(0|0)$ and an initial covariance matrix $\mathbf{P}(0|0)$; Let $k = 1$;
- (2) Use Eqs. (8) and (9) to obtain $\hat{\mathbf{x}}(k|k-1)$ and $\mathbf{P}(k|k-1)$, and use Cholesky decomposition to obtain $\mathbf{B}_p(k|k-1)$;
- (3) Let $t = 1$ and $\hat{\mathbf{x}}(k|k)_0 = \hat{\mathbf{x}}(k|k-1)$, where $\hat{\mathbf{x}}(k|k)_t$ denotes the estimated state at the fixed-point iteration t ;

(4) Use (26)–(32) to compute $\hat{\mathbf{x}}(k|k)_t$;

$$\hat{\mathbf{x}}(k|k)_t = \hat{\mathbf{x}}(k|k-1) + \tilde{\mathbf{K}}(k) (\mathbf{y}(k) - \mathbf{H}(k)\hat{\mathbf{x}}(k|k-1)) \quad (26)$$

with

$$\tilde{\mathbf{K}}(k) = \tilde{\mathbf{P}}(k|k-1)\mathbf{H}^T(k) (\mathbf{H}(k)\tilde{\mathbf{P}}(k|k-1)\mathbf{H}^T(k) + \tilde{\mathbf{R}}(k))^{-1}, \quad (27)$$

$$\tilde{\mathbf{P}}(k|k-1) = \mathbf{B}_p(k|k-1)\tilde{\mathbf{C}}_x^{-1}(k)\mathbf{B}_p^T(k|k-1), \quad (28)$$

$$\tilde{\mathbf{R}}(k) = \mathbf{B}_r(k)\tilde{\mathbf{C}}_y^{-1}(k)\mathbf{B}_r^T(k). \quad (29)$$

$$\tilde{\mathbf{C}}_x(k) = \text{diag} (G_\sigma (\tilde{e}_1(k)), \dots, G_\sigma (\tilde{e}_n(k))) \quad (30)$$

$$\tilde{\mathbf{C}}_y(k) = \text{diag} (G_\sigma (\tilde{e}_{n+1}(k)), \dots, G_\sigma (\tilde{e}_{n+m}(k))) \quad (31)$$

$$\tilde{e}_i(k) = d_i(k) - \mathbf{w}_i(k)\hat{\mathbf{x}}(k|k)_{t-1}. \quad (32)$$

(5) Compare the estimation of the current step and the estimation of the last step. If (33) holds, set $\hat{\mathbf{x}}(k|k) = \hat{\mathbf{x}}(k|k)_t$ and continue to (6). Otherwise, $t+1 \rightarrow t$, and go back to (4).

$$\frac{\|\hat{\mathbf{x}}(k|k)_t - \hat{\mathbf{x}}(k|k)_{t-1}\|}{\|\hat{\mathbf{x}}(k|k)_{t-1}\|} \leq \varepsilon. \quad (33)$$

(6) Update the posterior covariance matrix by (34), $k+1 \rightarrow k$ and go back to (2).

$$\mathbf{P}(k|k) = (\mathbf{I} - \tilde{\mathbf{K}}(k)\mathbf{H}(k)) \mathbf{P}(k|k-1) (\mathbf{I} - \tilde{\mathbf{K}}(k)\mathbf{H}(k))^T + \tilde{\mathbf{K}}(k)\mathbf{R}(k)\tilde{\mathbf{K}}^T(k) \quad (34)$$

Remark 2. As one can see, different from the traditional KF algorithm, the MCKF uses a fixed-point algorithm to update the posterior estimate of the state. The small positive number ε provides a stop condition (or a threshold) for the fixed-point iteration. Since the initial value of the fixed-point iteration is set at the prior estimate $\hat{\mathbf{x}}(k|k-1)$, the convergence to the optimal solution will be very fast (usually in several steps).

The kernel bandwidth σ is a key parameter in MCKF. In general, a smaller kernel bandwidth makes the algorithm more robust (with respect to outliers) but converge more slowly. On the other hand, when σ becomes more and more larger, the MCKF will behave more and more like the ordinary KF algorithm. In particular, the following theorem holds.

Theorem 3. When the kernel bandwidth $\sigma \rightarrow \infty$, the MCKF will reduce to the KF algorithm.

Proof. As $\sigma \rightarrow \infty$, we have $\mathbf{C}(k) \rightarrow \mathbf{I}$. In this case, the MCKF becomes the standard KF.

3.2. Computational complexity

Next, we analyze the computational complexity in terms of the floating point operations for the proposed algorithm. The computational complexities of some basic equations are given in Table 1.

The traditional Kalman filter algorithm involves Eqs. (8)–(12). Thus, from Table 1, one can conclude that the computational complexity of Kalman filter is

$$S_{KF} = 8n^3 + 10n^2m - n^2 + 6nm^2 - n + O(m^3). \quad (35)$$

The MCKF algorithm mainly involves Eqs. (8), (9), (26)–(32) and (34). Note that $\tilde{\mathbf{C}}_x(k)$ and $\tilde{\mathbf{C}}_y(k)$ are diagonal matrices, so it is very easy to obtain their inverse matrices. Assume that the average fixed-point iteration number is T . Then, according to Table 1, the computational complexity of the MCKF is

$$\begin{aligned} S_{MCKF} = & (2T+8)n^3 + (4T+6)n^2m + (2T-1)n^2 \\ & + (4T+2)nm^2 + (3T-1)nm + (4T-1)n \\ & + 2Tm^3 + 2Tm + TO(n^3) + 2TO(m^3). \end{aligned} \quad (36)$$

Table 1

Computational complexities of some equations.

Equation	Addition/subtraction and multiplication	Division, matrix inversion, Cholesky decomposition and exponentiation
(8)	$2n^2 - n$	0
(9)	$4n^3 - n^2$	0
(10)	$4n^2m + 4nm^2 - 3nm$	$O(m^3)$
(11)	$4nm$	0
(12)	$4n^3 + 6n^2m - 2n^2 + 2nm^2 - nm$	0
(26)	$4nm$	0
(27)	$4n^2m + 4nm^2 - 3nm$	$O(m^3)$
(28)	$2n^3$	$n + O(n^3)$
(29)	$2m^3$	$m + O(m^3)$
(30)	$2n^2$	n
(31)	$2nm$	m
(32)	$2n$	0
(34)	$4n^3 + 6n^2m - 2n^2 + 2nm^2 - nm$	0

The fixed-point iteration number T is relatively small in general (see the simulation results in Section 4). Thus the computational complexity of the MCKF is moderate compared with the traditional KF algorithm.

3.3. Convergence issue

The rigorous convergence analysis of the proposed MCKF algorithm is very complicated. In the following, we present only a sufficient condition that guarantees the convergence of the fixed-point iterations in MCKF. The result is similar to that of Chen et al. (2015) and hence, will not be proven here.

Let $\|\cdot\|_p$ denote an l_p -norm of a vector or an induced norm of a matrix defined by $\|\mathbf{A}\|_p = \max_{\|\mathbf{x}\|_p \neq 0} \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p}$ with $p \geq 1$, and $\lambda_{\min}[\cdot]$ denote the minimum eigenvalue of a matrix. According to the results of Chen et al. (2015), the following theorem holds.

Theorem 4. If $\beta > \zeta = \frac{\sqrt{n} \sum_{i=1}^L \|\mathbf{w}_i^T(k)\|_1 |d_i(k)|}{\lambda_{\min}[\sum_{i=1}^L \mathbf{w}_i^T(k)\mathbf{w}_i(k)]}$, and $\sigma \geq \max\{\sigma^*, \sigma^\dagger\}$, in which σ^* is the solution of the equation $\phi(\sigma) = \beta$, with

$$\begin{aligned} \phi(\sigma) = & \frac{\sqrt{n} \sum_{i=1}^L \|\mathbf{w}_i^T(k)\|_1 |d_i(k)|}{\lambda_{\min} \left[\sum_{i=1}^L G_\sigma(\beta \|\mathbf{w}_i(k)\|_1 + |d_i(k)|) \mathbf{w}_i^T(k)\mathbf{w}_i(k) \right]}, \\ & \sigma \in (0, \infty) \end{aligned} \quad (37)$$

and $\sigma^\dagger$ is the solution of equation $\psi(\sigma) = \alpha$ ($0 < \alpha < 1$), with Eq. (38) in Box 1. then it holds that $\|\mathbf{f}(\mathbf{x}(k))\|_1 \leq \beta$, and $\|\nabla_{\mathbf{x}(k)} \mathbf{f}(\mathbf{x}(k))\|_1 \leq \alpha$ for all $\mathbf{x}(k) \in \{\mathbf{x}(k) \in \mathbb{R}^n : \|\mathbf{x}(k)\|_1 \leq \beta\}$, where $\nabla_{\mathbf{x}(k)} \mathbf{f}(\mathbf{x}(k))$ denotes the $n \times n$ Jacobian matrix of $\mathbf{f}(\mathbf{x}(k))$ with respect to $\mathbf{x}(k)$, that is

$$\nabla_{\mathbf{x}(k)} \mathbf{f}(\mathbf{x}(k)) = \left[\frac{\partial}{\partial x_1(k)} \mathbf{f}(\mathbf{x}(k)) \dots \frac{\partial}{\partial x_n(k)} \mathbf{f}(\mathbf{x}(k)) \right] \quad (39)$$

with

$$\begin{aligned} & \frac{\partial}{\partial x_j(k)} \mathbf{f}(\mathbf{x}(k)) \\ &= -\mathbf{N}_{\mathbf{w}\mathbf{w}}^{-1} \left(\frac{1}{\sigma^2} \sum_{i=1}^L \left[e_i(k) w_i^j(k) G_\sigma(e_i(k)) \mathbf{w}_i^T(k) \mathbf{w}_i(k) \right] \right) \mathbf{f}(\mathbf{x}(k)) \\ &+ \mathbf{N}_{\mathbf{w}\mathbf{w}}^{-1} \left(\frac{1}{\sigma^2} \sum_{i=1}^L \left[e_i(k) w_i^j(k) G_\sigma(e_i(k)) \mathbf{w}_i^T(k) d_i(k) \right] \right) \end{aligned}$$

$$\psi(\sigma) = \frac{\sqrt{n} \sum_{i=1}^L [(\beta \|\mathbf{w}_i(k)\|_1 + |d_i(k)|) \|\mathbf{w}_i(k)\|_1 (\beta \|\mathbf{w}_i^T(k) \mathbf{w}_i(k)\|_1 + \|\mathbf{w}_i^T(k) d_i(k)\|_1)]}{\sigma^2 \lambda_{\min} \left[\sum_{i=1}^L G_{\sigma} (\beta \|\mathbf{w}_i(k)\|_1 + |d_i(k)|) \mathbf{w}_i^T(k) \mathbf{w}_i(k) \right]}, \quad \sigma \in (0, \infty) \quad (38)$$

Box I.

where $\mathbf{N}_{\mathbf{w}\mathbf{w}} = \sum_{i=1}^L G_{\sigma} (e_i(k)) \mathbf{w}_i^T(k) \mathbf{w}_i(k)$ and $w_i^j(k)$ is the j th element of $\mathbf{w}_i(k)$.

By Theorem 4, if the kernel bandwidth σ is larger than a certain value, we have

$$\begin{cases} \|\mathbf{f}(\mathbf{x}(k))\|_1 \leq \beta \\ \|\nabla_{\mathbf{x}(k)} \mathbf{f}(\mathbf{x}(k))\|_1 \leq \alpha < 1. \end{cases} \quad (40)$$

By Banach Fixed-Point Theorem (Agarwal et al., 2001), given an initial state estimate satisfying $\|\mathbf{x}(k)_0\|_1 \leq \beta$, the fixed-point iteration algorithm in MCKF will surely converge to a unique fixed point in the range $\mathbf{x}(k) \in \{\mathbf{x}(k) \in \mathbb{R}^n : \|\mathbf{x}(k)\|_1 \leq \beta\}$ provided that the kernel bandwidth σ is larger than a certain value (e.g. $\max\{\sigma^*, \sigma^\dagger\}$).

Theorem 4 implies that the kernel bandwidth has a significant influence on the convergence behavior of MCKF. If the kernel bandwidth σ is too small, the algorithm will diverge or converge very slowly. A larger kernel bandwidth ensures a faster converge speed but usually leads to a poor performance in impulsive noises. In practical applications, the kernel bandwidth can be set manually or optimized by trial and error methods.

4. Illustrative examples

In this section, we present some illustrative examples to demonstrate the performance of the proposed MCKF algorithm.

4.1. Example 1

Consider the following linear system:

$$\begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x_1(k-1) \\ x_2(k-1) \end{bmatrix} + \begin{bmatrix} q_1(k-1) \\ q_2(k-1) \end{bmatrix} \quad (41)$$

$$y(k) = [1 \quad 1] \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + r(k) \quad (42)$$

where $\theta = \pi/18$.

First, we consider the case in which the noises are all Gaussian, that is

$$q_1(k-1) \sim N(0, 0.01)$$

$$q_2(k-1) \sim N(0, 0.01)$$

$$r(k) \sim N(0, 0.01).$$

Table 2 shows the MSEs of x_1 and x_2 for different filters. Here, the MSE is computed as an average over 100 independent Monte Carlo runs, and in each run, 1000 samples (time steps) are used to evaluate the MSE. Since all the noises are Gaussian, the Kalman filter performs very well and in this example, it achieves almost the best performance (that is, the smallest MSEs). One can also see that when the kernel bandwidth is too small, the MCKF may achieve a worse performance; while when the kernel bandwidth becomes larger, its performance will approach that of the KF. Actually, it has been proven that when $\sigma \rightarrow \infty$, the MCKF will reduce to

Table 2
MSEs of x_1 and x_2 in Gaussian noises.

Filter	MSE of x_1	MSE of x_2
KF	0.035778	0.030052
MCKF ($\sigma = 0.5, \varepsilon = 10^{-6}$)	0.131361	0.105729
MCKF ($\sigma = 1.0, \varepsilon = 10^{-6}$)	0.103497	0.096126
MCKF ($\sigma = 3.0, \varepsilon = 10^{-6}$)	0.035885	0.030139
MCKF ($\sigma = 5.0, \varepsilon = 10^{-6}$)	0.035785	0.030047
MCKF ($\sigma = 7.0, \varepsilon = 10^{-6}$)	0.035784	0.030051

Table 3
Average iteration numbers for every time step with different σ .

Filter	Average iteration numbers
MCKF ($\sigma = 0.2, \varepsilon = 10^{-6}$)	3.89939
MCKF ($\sigma = 0.3, \varepsilon = 10^{-6}$)	3.43072
MCKF ($\sigma = 0.5, \varepsilon = 10^{-6}$)	2.78181
MCKF ($\sigma = 1.0, \varepsilon = 10^{-6}$)	2.36034
MCKF ($\sigma = 2.0, \varepsilon = 10^{-6}$)	2.12750
MCKF ($\sigma = 3.0, \varepsilon = 10^{-6}$)	2.01670
MCKF ($\sigma = 10, \varepsilon = 10^{-6}$)	1.67353

the traditional KF. In general, one should choose a larger kernel bandwidth under Gaussian noises.

Second, we consider the case in which the process noises are still Gaussian but the observation noise is a heavy-tailed (impulsive) non-Gaussian noise, with a mixed-Gaussian distribution, that is

$$q_1(k-1) \sim N(0, 0.01)$$

$$q_2(k-1) \sim N(0, 0.01)$$

$$r(k) \sim 0.9N(0, 0.01) + 0.1N(0, 100).$$

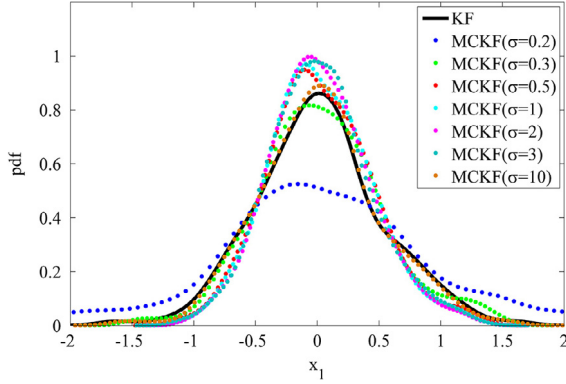
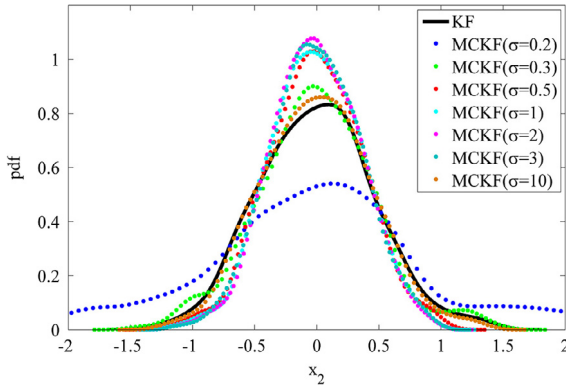
Figs. 1 and 2 illustrate the probability densities of the estimation errors of x_1 and x_2 . In the simulation, we set $\varepsilon = 10^{-6}$. As one can see, in impulsive noises, when kernel bandwidth is too small or too large, the performance of MCKF will be not good. In this case, however, with a proper kernel bandwidth (say $\sigma = 2.0$), the MCKF can outperform the KF significantly, achieving a desirable error distribution with a higher peak and smaller dispersion. Again, when σ is very large, MCKF achieves almost the same performance as the KF.

Table 3 shows the fixed-point iteration numbers for every time step with different kernel bandwidths, which are computed as averages over 100 independent Monte Carlo runs, with each run containing 1000 time steps. It is evident that the larger the kernel bandwidth, the faster the convergence speed. In particular, when the kernel bandwidth is large enough, the fixed-point algorithm in MCKF will converge to the optimal solution in just one or two iterations. In practical applications, to avoid slow convergence, the kernel bandwidth cannot be set at a very small value.

We further investigate the influence of the threshold ε on the performance. Table 4 illustrates the MSEs of x_1 and x_2 and average fixed-point iteration numbers with different ε (where the kernel bandwidth is set at $\sigma = 2.0$). One can see that a smaller ε usually results in slightly lower MSEs but needs more iterations to converge. Obviously, the influence of ε is not significant compared with the kernel bandwidth σ .

Table 4MSEs of x_1 and x_2 and average iteration numbers for every time step with different ε .

Filter	MSE of x_1	MSE of x_2	Average iteration numbers
KF	0.334496	0.280783	N/A
MCKF ($\sigma = 2.0, \varepsilon = 10^{-1}$)	0.221182	0.168721	1.03884
MCKF ($\sigma = 2.0, \varepsilon = 10^{-2}$)	0.220386	0.167958	1.09134
MCKF ($\sigma = 2.0, \varepsilon = 10^{-4}$)	0.220326	0.167900	1.44313
MCKF ($\sigma = 2.0, \varepsilon = 10^{-6}$)	0.220322	0.167899	2.13328
MCKF ($\sigma = 2.0, \varepsilon = 10^{-8}$)	0.220322	0.167899	2.78392

**Fig. 1.** Probability densities of x_1 estimation errors with different filters.**Fig. 2.** Probability densities of x_2 estimation errors with different filters.

4.2. Example 2

Now, we consider a practical example about one-dimensional linear uniformly accelerated motion. The state vector is $\mathbf{x}(k) = [x_1(k) \ x_2(k) \ x_3(k)]^T$, in which $x_1(k)$ is the position, $x_2(k)$ denotes the speed, and $x_3(k)$ stands for the acceleration. We assume that there are certain noises in the system and only the speed can be observed, which is also affected by some measurement disturbances. ΔT represents the measurement time interval. Then, the state and measurement equations are given by

$$\begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} = \begin{bmatrix} 1 & \Delta T & \Delta T^2/2 \\ 0 & 1 & \Delta T \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k-1) \\ x_2(k-1) \\ x_3(k-1) \end{bmatrix} + \begin{bmatrix} q_1(k-1) \\ q_2(k-1) \\ q_3(k-1) \end{bmatrix} \quad (43)$$

$$y(k) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + r(k) \quad (44)$$

with $\Delta T = 0.1$ s. We consider the situation where the process and measurement noises are all non-Gaussian with mixed-Gaussian

Table 5MSEs of x_1 , x_2 and x_3 in non-Gaussian process and measurement noises.

Filter	MSE of x_1 (m ²)	MSE of x_2 (m ² /s ²)	MSE of x_3 (m ² /s ⁴)
KF	155.3538	2.6866	2.1199
MCKF	78.1482	1.5068	1.8754
PF	537.6517	0.6687	8.8168

distributions, that is

$$q_1(k-1) \sim 0.9N(0, 0.01) + 0.1N(0, 1)$$

$$q_2(k-1) \sim 0.9N(0, 0.01) + 0.1N(0, 1)$$

$$q_3(k-1) \sim 0.9N(0, 0.01) + 0.1N(0, 1)$$

$$r(k) \sim 0.8N(0, 0.01) + 0.2N(0, 100).$$

Further, the initial values of the true state, estimated state and covariance matrix are assumed to be

$$\mathbf{x}(0) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T,$$

$$\hat{\mathbf{x}}(0|0) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T + N(0, 0.01) \times \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T,$$

$$\mathbf{P}(0|0) = 0.01 \times \text{diag}(1, 1, 1).$$

Table 5 summarizes the corresponding MSEs of x_1 , x_2 and x_3 for KF, MCKF and PF. In the simulation, the parameters are set at $\sigma = 2.0$, $\varepsilon = 10^{-6}$. The results confirm again that the proposed MCKF can outperform the traditional KF significantly when the system is disturbed by non-Gaussian process and measurement noises. Although the particle filter can achieve the smallest MSE in x_2 , it performs very poor when estimating x_1 and x_3 . One possible reason for this is that the PF may perform poor when some components cannot be measured directly (In this example, only the speed x_2 can be measured). High computational complexity of the particle filter also hinders its practical application.

5. Conclusion

A new Kalman type filtering algorithm, called maximum correntropy Kalman filter (MCKF), has been proposed in this work. The MCKF is derived by using the maximum correntropy criterion (MCC) as the optimality criterion, instead of using the well-known minimum mean square error (MMSE) criterion. The propagation equations for the prior estimates of the state and covariance matrix in MCKF are the same as those in KF. However, different from the KF, the MCKF uses a novel fixed-point algorithm to update the posterior estimations. The computational complexity of the MCKF is not expensive and the convergence is ensured if the kernel bandwidth is larger than a certain value. When the kernel bandwidth is large enough, the MCKF will behave like the KF. With a proper kernel bandwidth, the MCKF can outperform the KF significantly, especially when the underlying system is disturbed by some impulsive non-Gaussian noises. At present, we only consider the linear models, the extension to nonlinear models will be investigated in a future work.

Appendix. Derivation of the formula (24)

$$\begin{aligned} \mathbf{W}(k) &= \mathbf{B}^{-1}(k) \begin{bmatrix} \mathbf{I} \\ \mathbf{H}(k) \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{B}_p^{-1}(k|k-1) & 0 \\ 0 & \mathbf{B}_r^{-1}(k) \end{bmatrix} \begin{bmatrix} \mathbf{I} \\ \mathbf{H}(k) \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{B}_p^{-1}(k|k-1) \\ \mathbf{B}_r^{-1}(k) \mathbf{H}(k) \end{bmatrix} \end{aligned} \quad (\text{A.1})$$

$$\mathbf{C}(k) = \begin{bmatrix} \mathbf{C}_x(k) & 0 \\ 0 & \mathbf{C}_y(k) \end{bmatrix} \quad (\text{A.2})$$

$$\begin{aligned} \mathbf{D}(k) &= \mathbf{B}^{-1}(k) \begin{bmatrix} \hat{\mathbf{x}}(k|k-1) \\ \mathbf{y}(k) \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{B}_p^{-1}(k|k-1) \hat{\mathbf{x}}(k|k-1) \\ \mathbf{B}_r^{-1}(k) \mathbf{y}(k) \end{bmatrix}. \end{aligned} \quad (\text{A.3})$$

By (A.1) and (A.2), we have

$$\begin{aligned} &(\mathbf{W}^T(k) \mathbf{C}(k) \mathbf{W}(k))^{-1} \\ &= \left[(\mathbf{B}_p^{-1})^T \mathbf{C}_x \mathbf{B}_p^{-1} + \mathbf{H}^T (\mathbf{B}_r^{-1})^T \mathbf{C}_y \mathbf{B}_r^{-1} \mathbf{H} \right]^{-1} \end{aligned} \quad (\text{A.4})$$

where we denote $\mathbf{B}_p(k|k-1)$ by $\mathbf{B}_p$, $\mathbf{B}_r(k)$ by $\mathbf{B}_r$, $\mathbf{C}_x(k)$ by $\mathbf{C}_x$, and $\mathbf{C}_y(k)$ by $\mathbf{C}_y$ for simplicity.

Using the matrix inversion lemma with the identification:

$$\begin{aligned} (\mathbf{B}_p^{-1})^T \mathbf{C}_x \mathbf{B}_p^{-1} &\rightarrow \mathbf{A}, & \mathbf{H}^T &\rightarrow \mathbf{B}, \\ \mathbf{H} &\rightarrow \mathbf{C}, & (\mathbf{B}_r^{-1})^T \mathbf{C}_y \mathbf{B}_r^{-1} &\rightarrow \mathbf{D}. \end{aligned}$$

We arrive at

$$\begin{aligned} &(\mathbf{W}^T(k) \mathbf{C}(k) \mathbf{W}(k))^{-1} \\ &= (\mathbf{B}_p \mathbf{C}_x^{-1} \mathbf{B}_p^T - \mathbf{B}_p \mathbf{C}_x^{-1} \mathbf{B}_p^T \mathbf{H}^T \\ &\quad \times (\mathbf{B}_r \mathbf{C}_y^{-1} \mathbf{B}_r^T + \mathbf{H} \mathbf{B}_p \mathbf{C}_x^{-1} \mathbf{B}_p^T \mathbf{H}^T)^{-1} \mathbf{H} \mathbf{B}_p \mathbf{C}_x^{-1} \mathbf{B}_p^T). \end{aligned} \quad (\text{A.5})$$

Further, by (A.1)–(A.3), we derive

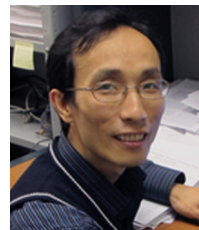
$$\begin{aligned} &\mathbf{W}^T(k) \mathbf{C}(k) \mathbf{D}(k) \\ &= (\mathbf{B}_p^{-1})^T \mathbf{C}_x \mathbf{B}_p^{-1} \hat{\mathbf{x}}(k|k-1) + \mathbf{H}^T (\mathbf{B}_r^{-1})^T \mathbf{C}_y \mathbf{B}_r^{-1} \mathbf{y}(k). \end{aligned} \quad (\text{A.6})$$

Combining (23), (A.5) and (A.6), we obtain (24).

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